

Adaptive basis set generation methods

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1 Introduction

The Density-Based Basis-Set Correction (DBBSC) method [?] is an electronic structure theory method based on the one-electron density. This method introduces a functional that corrects electron correlation and allows to achieve convergence to the complete basis set limit. The high accuracy of DBCBS makes it interesting for use in solving electronic structure theory problems on quantum simulators, for future quantum computer benchmarking purposes. We study numerical tools that make the implementation of this method possible on quantum simulators with a number of available qubits up to 28. This corresponds to a number of doubly occupied molecular orbitals [spin-orbitals?] equal to 14.

Existing molecular orbital discretisation methods include wavelets [?] and linear combinations of atomic orbitals, for instance Gaussian-type orbitals. We focus on the latter. Existing methods for compressing molecular orbitals, such as the active space method [?], do not easily allow to reduce the number of MOs to minimal sizes while respecting chemical accuracy. Other existing methods based on the Cholesky decomposition [?] of the Hamiltonian operator employ low rank approximations to improve storage requirements and operation count. Such class of methods is based on matrix or tensor compression from a linear algebra point of view. Here, we investigate alternative density-based solutions, that achieve low computational and storage cost by approximating the electronic density function using smaller basis sets. The advantage of this approach is that the resulting smaller basis set can be used for multiple purposes. Namely, it can directly be used as a discretisation basis for minimizing any given energy functional, since it is Hamiltonian-independent.

The main motivation behind our approach is the presence of linear dependencies, which are intrinsic to any given atomic orbital basis set of Gaussian-type orbitals. Inspired by density fitting methods [?], our strategy aims at detecting linear dependencies. This technique produces a smaller basis set using a density-based criterion, which is the approximation of a given target electron density under any given norm. In the present work, we propose new adaptive basis set generation methods based on density fitting. We divide the computation time into an *offline phase*, performed on a classical computer, and an *online phase*, performed on a quantum simulator. The calculation of the target electron density, used as a reference, and the adapted basis generation are both performed during the offline phase.

The present manuscript is organized as follows. The first part describes the mathematical background of our adapted basis set generation methods and the second part contains numerical results. The performance of our method is numerically assessed on small molecules.

2 Theory

We briefly describe the theory of the DBBSC method, originally proposed in [?]. Let us consider a system of N electrons and the electronic Hamiltonian

$$H = T + W_{ee} + V_{ne},$$

where T is the kinetic operator, W_{ee} is the electron repulsion operator, given by $\sum_{i < j}^N r_{ij}^{-1}$, and V_{ne} is the nuclei-electron attraction operator. The exact electronic ground state energy problem reads

$$E_0 = \min_{\Psi \in W} \langle \Psi, H \Psi \rangle, \quad (1)$$

where the exact wave function lives in the space defined as

$$W := \{\Psi \in \wedge^N H^1(\mathbb{R}^3, \mathbb{C}) : \langle \Psi, \Psi \rangle = 1\}.$$

We discretise problem of Eq. (1) by introducing a given finite basis set, denoted by $B \subset H^1(\mathbb{R}^3, \mathbb{C})$, and the discretised wave function space

$$W^B := \{\Psi \in \wedge^N \text{Span}(B) : \langle \Psi, \Psi \rangle = 1\}.$$

The discretised ground state problem then reads

$$E_*^B = \min_{\Psi \in W^B} \langle \Psi, H \Psi \rangle. \quad (2)$$

The limitation of estimating the exact ground state energy by solving the discretised problem of Eq. (2) is that the error on the energy converges as $\mathcal{O}(|B|^{-1})$, due to the cusp of the exact electron wave function. In order to improve the error convergence, the DBBSC method proposes to approximate the ground state energy by solving the following discretised problem

$$E_0^B = \min_{\Psi \in W^B} (\langle \Psi, H \Psi \rangle + \bar{E}^B[n_\Psi]), \quad (3)$$

with $\bar{E}^B[n_\Psi]$ the basis set correction functional, we refer to the original work for its explicit definition. The one-electron density n_Ψ , for any given wave function $\Psi \in W^B$, lives in the space

$$D^B := \{n \in D : \exists \Psi \in W^B, \quad n_\Psi = n\},$$

with

$$D = \{n \in L^1(\mathbb{R}^3) : n \geq 0, \quad \int_{\mathbb{R}^3} n = N, \quad \sqrt{n} \in H^1(\mathbb{R}^3)\}.$$

The problem of Eq. (3) is solved self-consistently [?].

2.1 Slater determinants

In Hartree-Fock and post Hartree-Fock methods, a special form of wave function is considered, known as a *Slater determinant*, defined as

$$\Psi(\mathbf{r}_1, \dots, \mathbf{r}_N) = \frac{1}{\sqrt{N!}} \det(\phi_i(\mathbf{r}_j))_{1 \leq i, j \leq N} \quad (4)$$

where each ϕ_i is a normalised one-electron function of $H^1(\mathbb{R}^3, \mathbb{C})$ referred to as Molecular Orbital (MO). It can be easily verified that Ψ in the form of Eq. (4) is indeed an element of W . For any given Slater determinant $\Psi \in W$, the one-electron density is given in terms of the Molecular Orbitals (MOs) as

$$n_\Psi = \sum_{i=1}^N |\phi_i|^2. \quad (5)$$

2.2 Approximating the density

Under the Linear Combination of Atomic Orbitals (LCAO) approximation, the discretisation basis B is denoted by $B = \{\chi_\mu\}_{1 \leq \mu \leq N_b}$, where each χ_μ is an atomic orbital basis function. Such discretisation yields an approximation of the exact MOs on the basis of AOs. The problem unknown is the coefficient matrix $\mathbf{C} \in \mathbb{R}^{N \times N_b}$ in the decomposition

$$\forall 1 \leq i \leq N, \quad \phi_i = \sum_{\mu=1}^{N_b} C_{i\mu} \chi_\mu.$$

Substituting this MO expression in the one-electron density of Eq. (5), one obtains the following expansion of the density on the AOs:

$$n_\Psi = \sum_{\mu=1}^{N_b} \sum_{\nu=1}^{N_b} D_{\mu\nu} \chi_\mu \chi_\nu, \quad (6)$$

where $\mathbf{D} = \mathbf{C}^\top \mathbf{C}$ is the so-called *density matrix*, of dimension $N_b \times N_b$.

2.3 Problem formulation

The problem consists in constructing a subset $I \subseteq \{1, \dots, N_b\}$ of any given size M such that if the restricted basis $B_I := \{\chi_\mu\}_{\mu \in I}$ is used to discretise the problem of Eq. (3), i.e. if the minimisation space is W^{B_I} instead of W^B , then the accuracy loss on the estimated ground state energy with respect to using the full basis B , i.e. the error

$$|E_0^B - E_0^{B_I}|,$$

remains small.

Our main idea for solving this problem can be described as follows. Let us denote by $\Psi_0 \in W^B$ the minimiser of problem of Eq. (3) and by $n_{\Psi_0} \in D^B$ the associated electron density. Given a target size M , we seek the optimal subset of B of the form B_I with $|I| = M$, obtained by solving the following optimisation problem

$$\min_{\substack{I \subseteq \{1, \dots, N_b\} \\ |I|=M}} \min_{m \in D^{B_I}} \|n_{\Psi_0} - m\|,$$

where $\|\cdot\|$ is any given norm. Note that the optimisation criterion we used is the orthogonal projection error (also known as best approximation error) with respect to the reference electron density. In other words, the problem boils down to selecting the index subset of size M such that the best approximation error is minimised.

Due to the expression of the density given by Eq. (6), any function $m \in D^{B_I}$ is of the form

$$m = \sum_{\mu \in I} \sum_{\nu \in I} C_{\mu\nu} \chi_\mu \chi_\nu.$$

2.4 Algorithms

To summarize, the main ingredients given as an input to our algorithm is

- given molecular geometry,
- atomic orbital basis set,
- a starting density matrix.

We studied three different choices of density matrices: Hartree-Fock (HF), Configuration Interaction Singles and Doubles (CISD) and Configuration Interaction Perturbatively Selected Iteratively (CIPSI).

2.4.1 Assembly of the Gram matrix

The first step is the assembly of the Gram matrix (i.e. the overlap matrix) of the orbital products, given a user-defined inner product $\langle \cdot, \cdot \rangle$. Let us denote by $\mathbf{T}$ the tensor of dimension $N_b \times N_b \times N_b \times N_b$, with entries defined as

$$T_{\mu\nu\kappa\lambda} = D_{\mu\nu} D_{\kappa\lambda} \langle \chi_\mu \chi_\nu, \chi_\kappa \chi_\lambda \rangle.$$

The Gram matrix $\mathbf{G}$, of dimension $N_b^2 \times N_b^2$ is then formed by folding pairwise the first two and the last two dimensions of the tensor $\mathbf{T}$. Then we also regroup by orbital types. Another option for initialising the Gram matrix consists in $\mathbf{R}^\top \mathbf{G} \mathbf{R}$, where $\mathbf{R}$ is the atomic restriction matrix that discards orbital products centered on midbonds.

2.4.2 Selecting orbital products

We seek a selection of orbital products of target size $M_* \ll N_b^2$ that best approximate the one-electron density. This selection can be performed using a criterion based on minimizing the orthogonal projection error with respect to the full space spanned by orbital products, in which the density belongs to. Orbital products are discarded using a pivoted Cholesky decomposition performed on the Gram matrix $\mathbf{G}$, also known as Gram-Schmidt procedure with pivoting. The orthonormalization procedure terminates when the orthogonal projection error is smaller than a given threshold. Note that the entries of the Gram matrix are weighted by the density matrix. Therefore, at the first iteration of the pivoted Cholesky decomposition, the selected pivot is influenced by the value of the density matrix.

The pivoted Cholesky decomposition under a given tolerance value $\varepsilon > 0$, performed on the matrix $\mathbf{G}$, yields an approximate rank $M \leq N_b$ of $\mathbf{G}$ and a selection of M rows (and columns) that approximately span the full row space under the tolerance value. We propose to use the atomic orbitals corresponding to the selected indices to build the set of adaptive basis functions. This strategy leads to our algorithm named ABS-0, for Adaptive Basis Set generation.

Additionally, it may be of interest to impose maximal orbital type constraints to the adaptive basis set, in favour of computational efficiency within the integral evaluation procedure. The preliminary treatment of the Gram matrix $\mathbf{G}$ that enables this constraint can be presented as follows. The main idea is to construct an intermediary atomic orbital basis respecting orbital type constraints. We propose two possible methods. For each $1 \leq \mu \leq N_b$, if the atomic orbital χ_μ has orbital type strictly higher than d -type, then

- (a) ABS-1: discard χ_μ .
- (b) ABS-2: for each orbital type $t \in \{s, p, d\}$, create a new atomic orbital having the orbital exponent and atomic center of χ_μ , but orbital type t .

It is assumed that all atomic basis functions are normalized, so that the diagonal of $\mathbf{G}$ consists of one entries.

3 Numerical experiments

3.1 Implementation details

The quantum simulator uses the Hamiltonian in the second quantisation formulation, that reads

$$\hat{H} = \sum_{\sigma} \sum_{ij}^{N_{\text{MO}}} h_{ij} \hat{a}_{i\sigma}^\dagger \hat{a}_{j\sigma} + \sum_{ijkl}^{N_{\text{MO}}} h_{ijkl} \hat{a}_{i\sigma}^\dagger \hat{a}_{k\sigma}^\dagger \hat{a}_{j\sigma} \hat{a}_{l\sigma}, \quad (7)$$

where N_{MO} is the number of basis functions used to discretise the molecular orbitals. Note that our method allows to reduce the number of MOs in this expression.

3.2 Preprint

The task of constructing basis sets of minimal size that achieve a target level of accuracy is treated as follows. In the present work, we propose to formulate our goal in terms of a constraint optimisation problem. Let us denote by $\Psi_0 \in W^B$ the wave function corresponding to the ground state energy, equal to a normalised Slater determinant associated to molecular orbitals that are approximated on a basis of Gaussian-type atomic orbitals (GTOs), denoted by $\mathcal{B} = \{\chi_\mu\}_{1 \leq \mu \leq N_b}$. We assume that the coefficients on the GTOs are obtained by a converged minimisation procedure. Let us denote the one-electron density associated to the ground state by n_0^B . Given a target basis size M , we seek the optimal subset of $\mathcal{B}$, of the form $\mathcal{B}_I = \{\chi_\mu\}_{\mu \in I}$ associated to a set of indices $I \subseteq \{1, \dots, N_b\}$ and $|I| = M$, where $|\cdot|$ denotes the cardinal of the set I , that minimises the error of the *best approximation* of the density n_0^B over the space of $\mathcal{B}_I$ -representable one-electron densities. The optimal subset of $\mathcal{B}$ can be found by solving the following optimisation problem for obtaining the corresponding index set I_* ,

$$I_* := \arg \min_{\substack{I \subseteq \{1, \dots, N_b\} \\ |I| = M}} \min_{m^{\mathcal{B}_I}} \|n_0^B - m\|, \quad (8)$$

where $\|\cdot\|$ is any given norm defined for functions over $\mathcal{R}^3$. In the present work, we consider the Coulomb norm, defined as $\|\cdot\| = \langle \cdot, \cdot \rangle$ with $\langle n, m \rangle := \iint_{\mathbb{R}^3 \times \mathbb{R}^3} \frac{n(r)m(r')}{|r-r'|} dr dr'$ associated to the space Sobolev etc (see ref). Let us note that ?? requires knowledge of the density n_0^B used as a reference, which is assumed to be obtained by performing precomputation on a classical computer.

Let us describe our main approach for solving the problem of ?. Our main goal is to discard GTO indices using a greedy procedure for eliminating linear dependencies in the basis spanning the one-electron density, i.e.

$$n_0^B \in \text{Span}(\chi_\mu \chi_\nu)_{1 \leq \mu \leq \nu \leq N_b}.$$

The idea of selecting AO products has been initially proposed in ref in the context of constructing auxiliary basis sets used in density fitting, in the aim of accelerating the computation of bielectronic integrals. In the present work, we propose a new scheme based on the linear dependence elimination for solving problem of ?, named Molecule-Adapted Basis Set (MABS) generation procedure. Let us assume that the density given as a reference writes as

$$n_0^B = \sum_{\mu=1}^{N_b} \sum_{\nu=1}^{N_b} D_\mu D_\nu \chi_\mu \chi_\nu,$$

where $D = CC^\top$ denotes the density matrix and C are the coefficients in the linear combination of atomic orbitals, obtained by a converged energy minimisation procedure. Our scheme is formulated in terms of matrices, as follows. We first define the four-index tensor T with entries the AO products weighted by the density matrix, i.e. $T_{\mu\nu\kappa\lambda} = D_\mu D_\nu \langle \chi_\mu \chi_\nu, \chi_\lambda \chi_\kappa \rangle$. The Gram matrix of the weighted AO products is then formed as follows. It is known in the literature that the set of AO products suffers from linear dependencies. The Pivoted Cholesky Decomposition (PCD) (ref Susi) is a known tool for tackling this problem and removing linear dependencies using an iterative procedure, that may be interpreted as a modified Gram-Schmidt orthonormalisation that discards functions that do not contribute to the full span of the space up to a tolerance threshold value. Before selecting indices, we perform a pre-processing step of the matrix by blocks, that is necessary in order to respect the orbital type of the desired elements in $\mathcal{B}$. Due to orbital symmetry, all orbital components belonging to the same type will be contracted, i.e.

$$\hat{T}_{\mu\nu\kappa\lambda} = \sum_{\alpha\beta\gamma\delta} P_{\alpha\mu} P_{\beta\nu} T_{\mu\nu\kappa\lambda}.$$

where P is defined as (...). This yields the contraction matrix M . Additionally, we eliminate orbital products not centered on atoms, using a restriction matrix. Then we apply PCD to this matrix using the machine epsilon as a target tolerance value. This gives a set of indices sorted in pivot-descending order. The AO product selected index set is then defined as $I_M := \{(\mu, \nu) : \}$ the M -first indices of PCD. Finally, we recover the AO index set as $I_0 = \{\mu : \exists \nu, (\mu, \nu) \in I_M \text{ or } (\nu, \mu) \in I_M\}$. A final post-processing step is applied in order to assure the element-wise consistency, according to which all elements of the same type are assigned the same AO basis set. The AO basis set associated to any element is then the union of indices that belong to the set I_0 . This yields the solution I_* to the problem. The MABS basis, solution to this problem, is defined as $\mathcal{B}_{I_*}$. Note that our MABS generation scheme essentially controls the size M of the selected products. The actual size of the AO basis set, equal to $|I_*|$ is only implicitly controlled during this procedure. As we will see in practice, the difference between $|I_*|$ and M typically does not vary more than 10.

3.3 Results

The first experiment allows to compare the results obtained from a DFT/B3LYP calculation on the water molecule at equilibrium geometry, using state-of-the-art auxiliary basis sets (RI and JKFIT) and our adaptive basis sets, obtained with methods ABS- x for $x = 0, 1, 2$. The aim of this comparison is to study which method achieves better accuracy with less basis functions. The approximation error is measured on ground state energies using the full atomic orbital basis set as a reference calculation. We are also interested in assessing the convergence of the approximation error of our method for varying tolerance values in the Cholesky decomposition.

4 Perspectives

Our method can be combined with orbital exponent and contraction coefficient optimization methods, namely with the one proposed in [?]. A possible direction that allows to combine both methods is to first apply our method for basis set size reduction, then apply their method on our resulting adapted basis set for further optimization, in the goal of increasing the accuracy using a fixed number of MOs.

References

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